

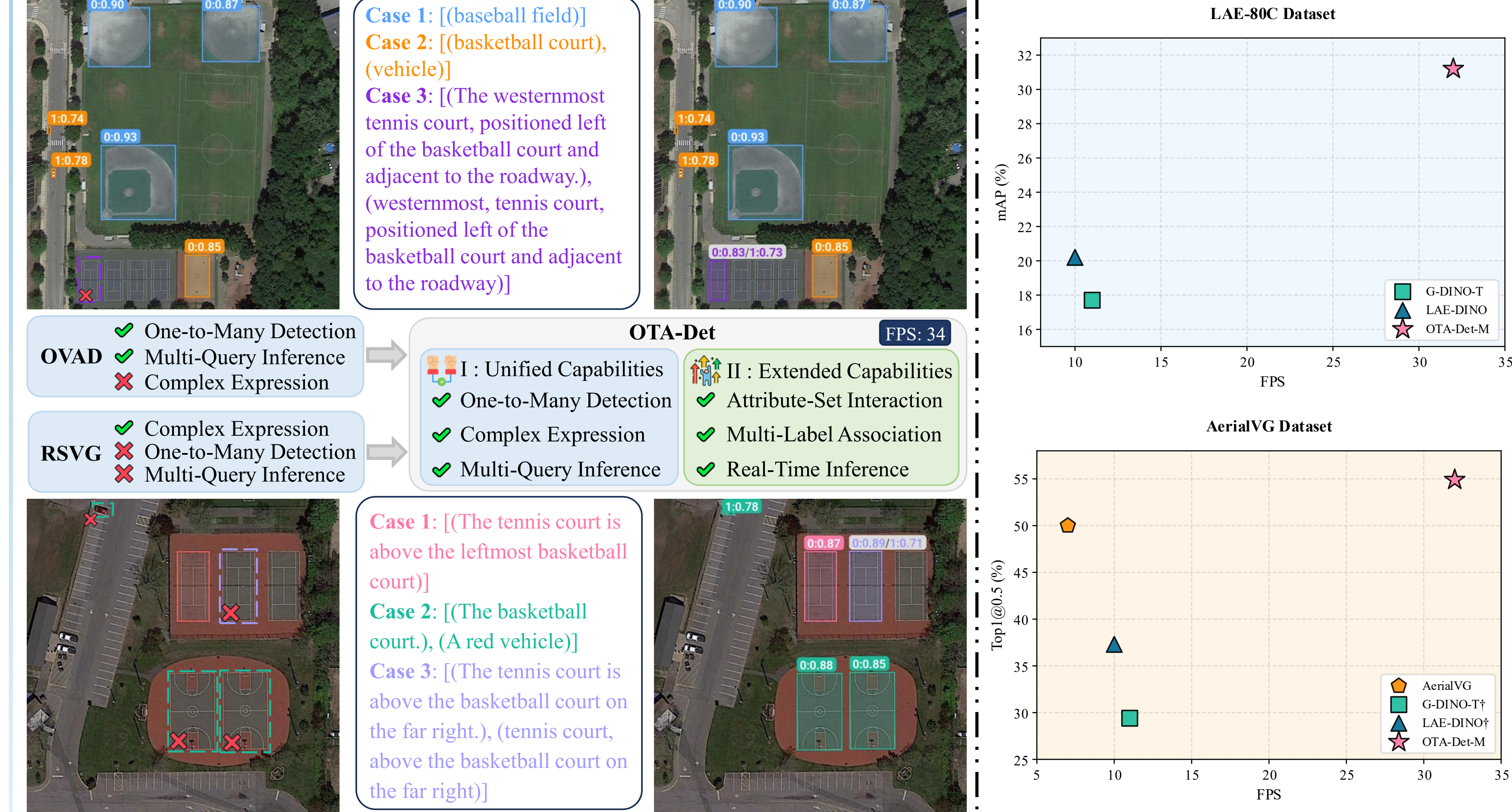
Open-Text Aerial Detection: A Unified Framework for Aerial Visual Grounding and Detection

Guoting Wei^{1,4,*}, Xia Yuan^{1,*}, Yang Zhou², Haizhao Jing², Yu Liu³, Xianbiao Qi⁴, Chunxia Zhao¹, Haokui Zhang^{2,4,†}, Rong Xiao⁴

¹Nanjing University of Science and Technology ²Northwestern Polytechnical University ³Zhejiang Lab ⁴Intellifusion

*Equal contribution †Corresponding author ✉ weiguoting@njust.edu.cn, hkzhang@nwpu.edu.cn WeChat: GuotingWei418

1. Motivation



Two existing paradigms suffer from inherent limitations when operating in isolation:

- OVAD:** detects *all* objects matching given category names, but only *coarse category-level* semantics, unable to parse complex descriptions.
- RSVG:** grounds *complex expressions*, but is restricted to a *single* target with *no category label*.

Real-world aerial perception needs *both* rich language understanding and multi-target detection.

2. Problem Formulation

Closed-set detection is limited to a *fixed* category set:

$$I \rightarrow \{(b_i, c_i, s_i)\}, \quad c_i \in \mathcal{C}_{\text{fix}}$$

OVAD extends it to *arbitrary* categories via vision-language alignment, yet stays *category-level*:

$$(I, \mathcal{T}_{\text{cat}}) \rightarrow \{(b_i, c_i, s_i)\}, \quad c_i \in \mathcal{T}_{\text{cat}}$$

RSVG parses a *complex* expression E but returns a *single, unlabeled* box:

$$(I, E) \rightarrow \{b\}$$

Two complementary halves, yet incompatible:

- Task gap.** OVAD jointly *classifies and localizes* all instances; RSVG is *localization-only*, returning one box with *no class label*.
- Alignment gap.** Both align text as a *holistic whole*, risking **semantic pseudo-alignment**: binding a sentence without grounding its *category, color, position*.

Our open-text formulation closes both gaps:

$$(I, \mathcal{T}_q, \mathcal{A}_q) \longrightarrow \{(b_i, c_i, s_i, \mathbf{c}_i^{\text{attr}}, \mathbf{s}_i^{\text{attr}})\}_{i=1}^N$$

b_i : box; (c_i, s_i) : query-level class & score; $(\mathbf{c}_i^{\text{attr}}, \mathbf{s}_i^{\text{attr}})$: attribute-level class & scores
 $\mathcal{T}_q = \mathcal{T}_{\text{cat}} \cup \{E_1, \dots, E_n\}$: categories and expressions; $\mathcal{A}_q$: decomposed attributes

3. Contributions

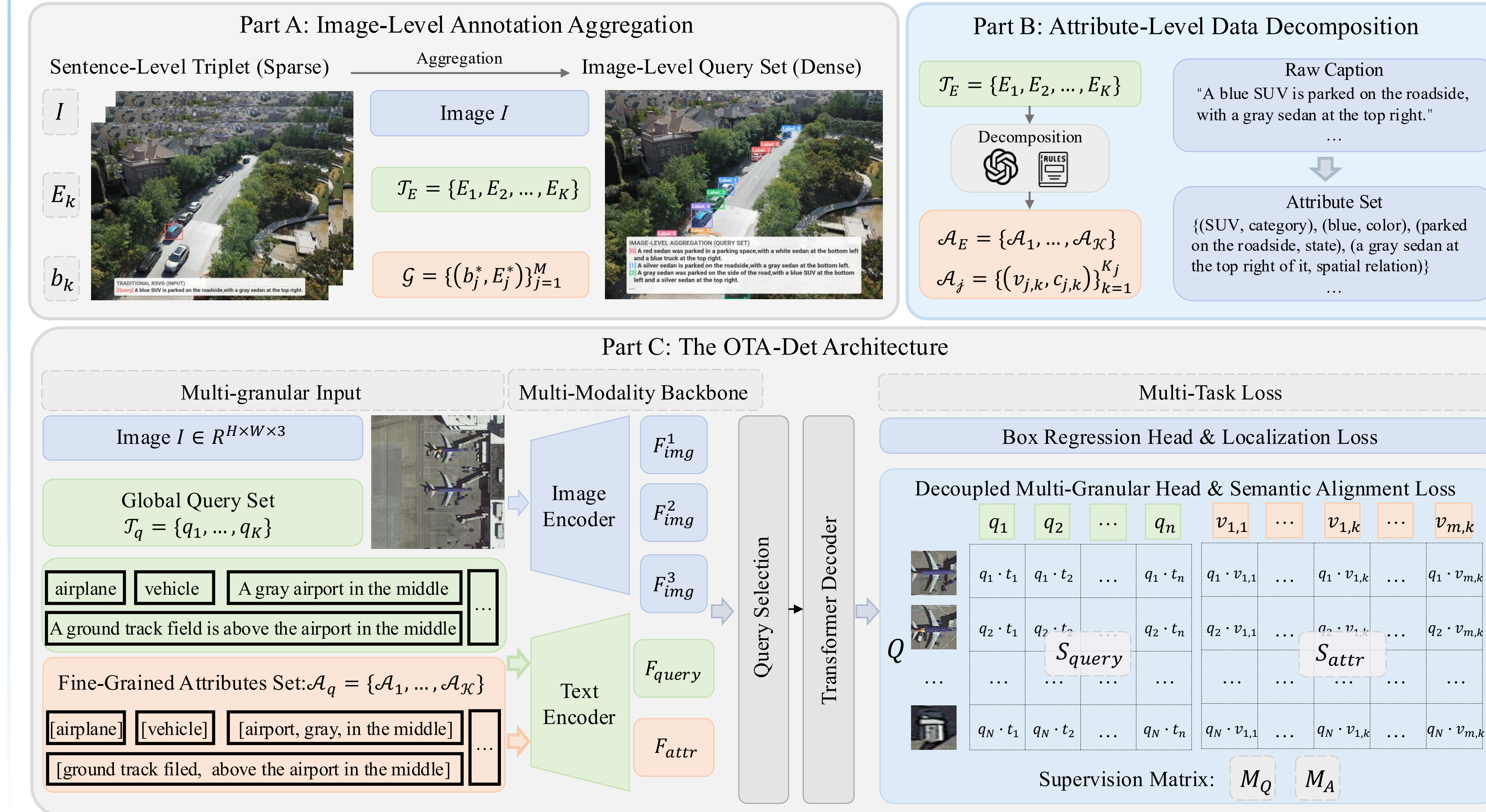
We propose **OTA-Det**, the *first* unified framework bridging OVAD and RSVG.

Task Reformulation Strategy (*closes the task gap*). Reformulates RSVG from pure localization into *joint classification-localization* aligned with OVAD; **Image-Level Annotation Aggregation** (**Part A**) adds **dense** supervision for **joint training** across paradigms.

Dense Semantic Alignment Strategy (*closes the alignment gap*). **Attribute-Level Data Decomposition** + **Unified Correspondence Matrix** + **Decoupled Multi-Granular Head** (**Parts B-C**) align text at *multiple granularities*, mitigating *pseudo-alignment* and enabling **attribute-set interaction**.

Real-Time Unified Architecture. Built on **RT-DETR** (DEIMv2), extending *closed-set* to *open-text* detection; **SOTA on 6 benchmarks** at **34 FPS**.

4. Method: OTA-Det Framework



5. Experiments & Results

Joint training on OTA-Mix (LAE-1M + OPT-RSVG + DIOR-RSVG + AerialVG)

METHOD	PRE-TRAIN	FPS	OVAD BENCHMARK			RSVG BENCHMARK		
			DIOR (AP ₅₀)	DOTA (MAP)	LAE-80C (MAP)	OPT-RSVG (Acc@0.5)	DIOR-RSVG (Acc@0.5)	AERIALVG (Acc@0.5)
<i>Open Vocabulary Aerial Detection</i>								
GLIP-T (CVPR'22)	LAE-1M	-	82.8	43.0	16.5	-	-	-
G-DINO-T (ECCV'24)	LAE-1M	11	83.6	46.0	17.7	-	-	-
LAE-DINO (AAAI'25)	LAE-1M	10	85.5	46.8	20.2	-	-	-
<i>Remote Sensing Visual Grounding</i>								
TRANSVG (ICCV'21)	-	-	-	-	-	70.0	72.4	11.5
MGVLF (TGRS'23)	-	-	-	-	-	72.2	76.8	-
LPVA (TGRS'24)	-	26	-	-	-	78.0	82.3	-
AERIALVG (ICCV'25)	-	7	-	-	-	-	-	50.0
<i>Unified Detectors</i>								
G-DINO-T [†] (ECCV'24)	OTA-MIX	11	89.8	45.3	23.7	<u>85.3</u>	<u>85.9</u>	27.5
LAE-DINO [†] (AAAI'25)	OTA-MIX	10	89.0	46.2	23.3	84.9	86.7	35.7
OTA-DET-S (OURS)	OTA-MIX	34	<u>91.9</u>	48.0	<u>29.2</u>	85.0	82.7	49.2
OTA-DET-M (OURS)	OTA-MIX	<u>32</u>	91.8	<u>49.4</u>	28.8	85.2	83.8	<u>51.7</u>
OTA-DET-L (OURS)	OTA-MIX	28	93.1	52.0	31.0	86.5	85.1	53.9

Separate per-task training & component ablation. (**Attr-Align@ τ** : a prediction is correctly localized (Acc@0.5) and its target's visual feature matches each attribute text with similarity $> \tau$.)

METHOD	DIOR	DOTA	LAE-80C
GLIP-T	82.8	43.0	16.5
G-DINO-T	83.6	46.0	17.7
LAE-DINO	85.5	46.8	20.2
OTA-DET-M	87.1	51.4	31.2

Separate Training (LAE-1M)

METHOD	Acc@0.5
TRANSVG	11.5
G-DINO [†]	29.4
LAE-DINO [†]	37.3
AERIALVG	50.0
OTA-DET-M	54.9

Separate Training: (AerialVG)

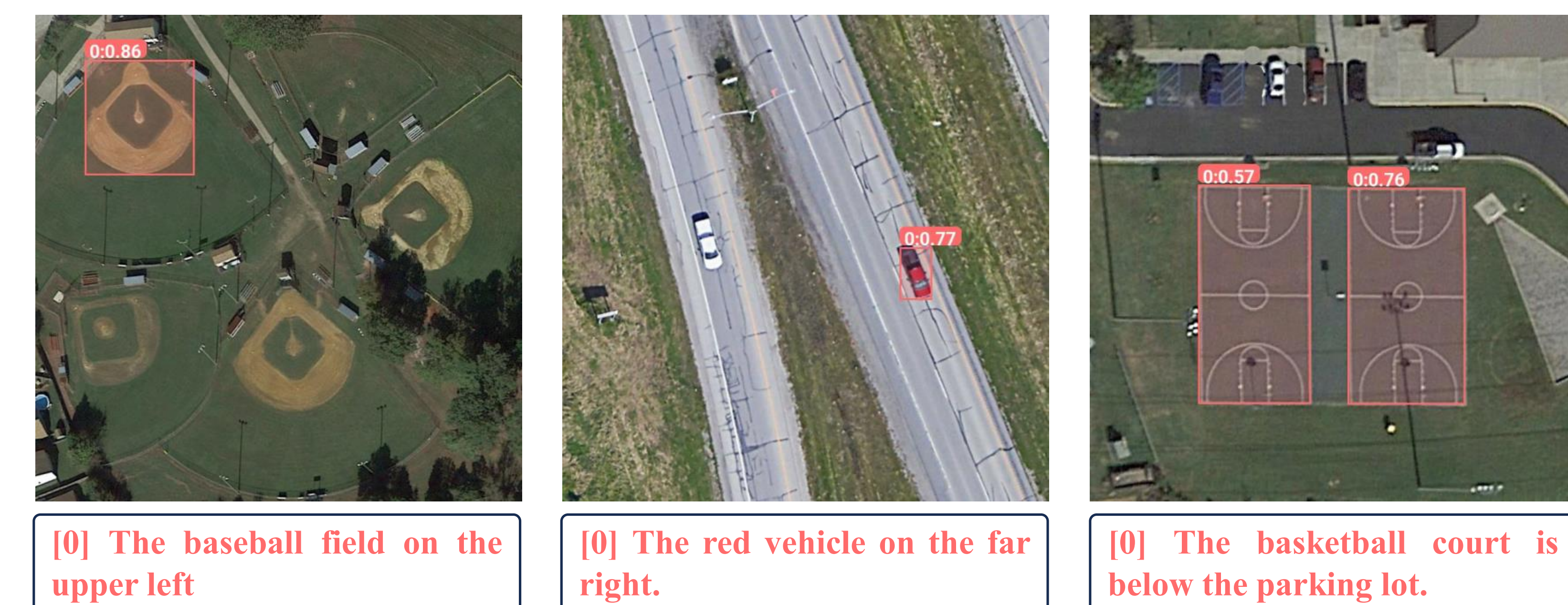
METHOD	ATTR-ALIGN $\uparrow$			
	STANDARD $\uparrow$ Acc@0.5	$\tau=0.5$	$\tau=0.6$	$\tau=0.7$
OTA-DET-M	54.94	44.23	34.36	21.96
w/o IMG-AGG.	54.56	43.21	33.90	22.74
w/o DECOUPLED	54.16	39.55	27.14	14.88
w/o DENSE ALIGN	54.67	14.36	3.81	0.47

Ablation: (AerialVG, OTA-Det-M)

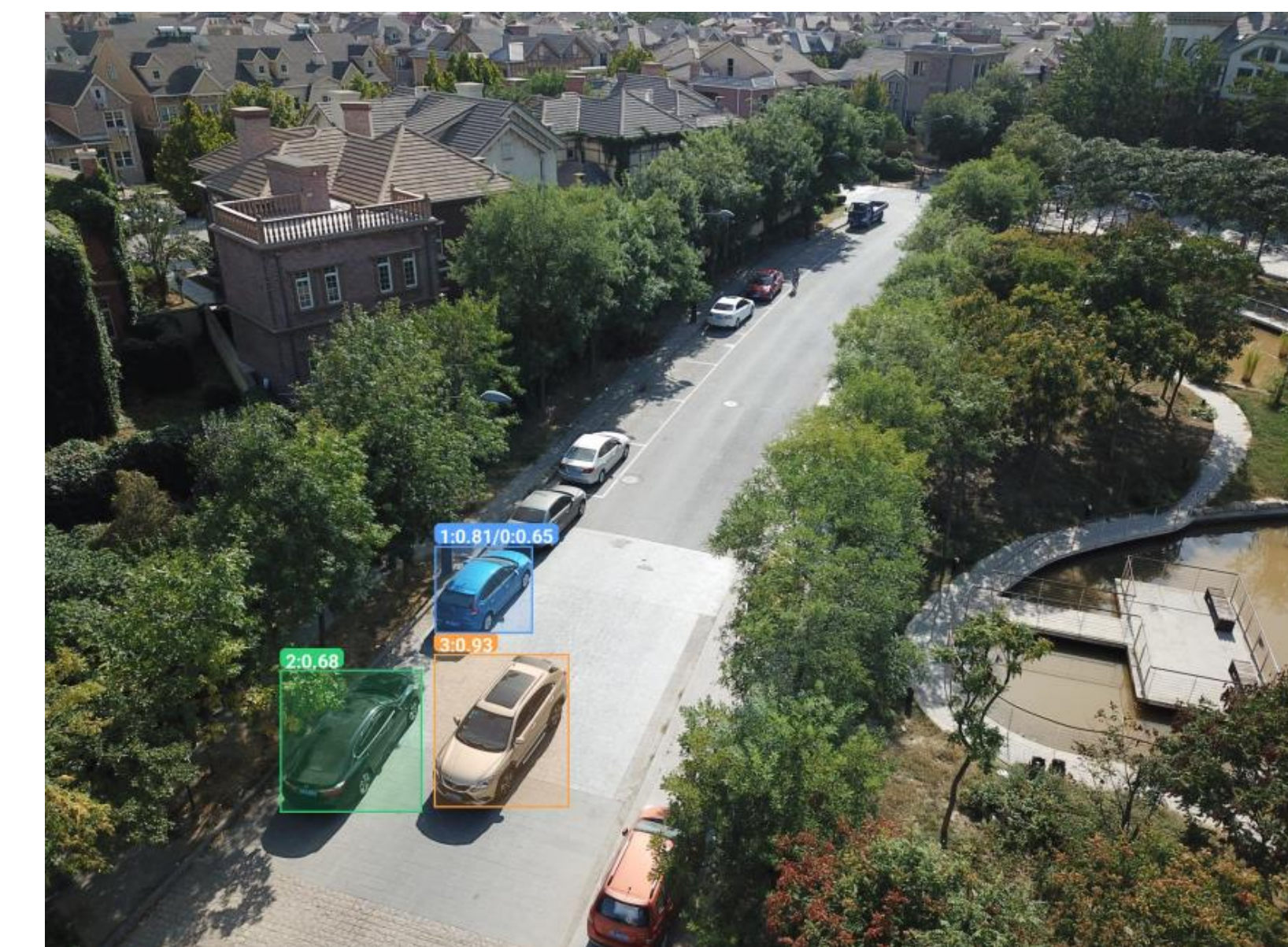
6. Qualitative Results



OVAD: multi-query, one-to-many detection in dense scenes



- [0]** [SUV, blue, is parked on the roadside, a gray sedan at the top right]
- [1]** A blue SUV is parked on the roadside, with a gray sedan at the top right.
- [2]** A black sedan is parked on the roadside, with a champagne-colored SUV on the right.
- [3]** A champagne-colored SUV was driving in the middle of the road, with an orange SUV at the bottom right and a black sedan on the left.



Grounding: spatial expressions, attribute-set, multi-label